

ADSP 31021 Machine Learning Operations

Assignment #4 - Model Monitoring

Due Date - Beginning of Session #8

Assignment Overview

The goal of this assignment is to provide hands-on experience building and monitoring a machine learning model after deployment-oriented evaluation. Students will develop a regression model using the attached cancer dataset, evaluate model performance, create controlled input shifts, and use model monitoring to detect changes in the input data and model output.

This assignment emphasizes model monitoring, data drift detection, prediction drift, and operational thinking. Students will compare baseline test performance with performance under controlled data changes and explain how monitoring helps identify risks before they affect production decisions.

Instructions

- Work individually unless otherwise approved.
- Clearly label all sections and tasks in your notebook or scripts.
- Use the attached cancer dataset and the provided Data Dictionary.docx for column definitions.
- Develop a machine learning model to predict target_deathrate using a train/test split methodology.
- Set up model monitoring using Evidently AI: <https://www.evidentlyai.com/>.
- Evaluate the model using the original test dataset and three modified test dataset scenarios.
- Your code must run end-to-end without manual intervention.
- Submission: Please upload your completed solution on Canvas or the submission mechanism specified by the instructor.
- Total Points: 100

Reproducibility Requirements

- All code must be executable from a clean environment.
- Include dependency management files such as requirements.txt or environment.yml.
- Use fixed random seed(s) for train/test split and model training where applicable.
- Clearly document execution steps for data loading, training, evaluation, perturbation, and monitoring.
- Preserve the original test dataset and create separate modified test datasets for each monitoring scenario.
- Save model evaluation metrics, monitoring reports, visualizations, and generated HTML reports where applicable.
- Document assumptions, feature handling decisions, and any columns excluded from modeling.

Required Tools

- Python 3.x
- Jupyter Notebook or Python scripts
- Pandas, NumPy, Scikit-learn, and visualization libraries as needed
- Evidently AI for model monitoring and drift reporting

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- Git and GitHub or another approved submission mechanism
- Libraries required for model development, evaluation, and report generation

Dataset

- Use the attached cancer dataset and the corresponding Data Dictionary.docx.
- Target variable: target_deathrate.
- The prediction task should be treated as a supervised regression problem. Students should use the data dictionary to understand feature meaning, identify potential data quality issues, and document any preprocessing assumptions.

Monitoring Scenarios

After evaluating the model on the original test dataset, create the following modified test datasets. Apply each change one at a time as specified below:

- Scenario A: Change medianincome by decreasing it by 40,000. Example: 43,823 becomes 3,823.
- Scenario A + B: Apply Scenario A, and also change povertypercent by increasing it by 20 points. Example: 11.9 becomes 31.9.
- Scenario A + B + C: Apply Scenario A and B, and also change avghouseholdsize by increasing it by 2. Example: 2.31 becomes 4.31.

For each scenario, run the model, verify the accuracy, and use model monitoring to detect changes in the input data and model output.

Required Workflow Tasks

1. Dataset Loading and Data Understanding (10 Points)

- Load the attached cancer dataset.
- Review the Data Dictionary.docx and summarize the meaning of key columns used in modeling.
- Identify the target column target_deathrate and document the prediction objective.
- Perform basic data validation, including column types, missing values, duplicate records, and obvious invalid values.
- Document assumptions and any columns excluded from the model.

2. Preprocessing and Train/Test Split (10 Points)

- Prepare the dataset for supervised regression modeling.
- Handle missing values, categorical variables, scaling, or transformations as appropriate.
- Create a train/test split with a fixed random seed.
- Document the split ratio, random seed, target variable, selected features, and preprocessing steps.
- Ensure preprocessing is fit only on training data where applicable to avoid data leakage.

3. Baseline Model Development (15 Points)

- Train a machine learning regression model to predict target_deathrate.

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- You may use Linear Regression, Random Forest, XGBoost, Gradient Boosting, Neural Networks, or another approved regression approach.
- Clearly document the selected model type, feature set, training methodology, and assumptions.
- Save the trained model or ensure the training pipeline can reproduce the model.

4. Baseline Model Evaluation (10 Points)

- Evaluate the trained model using the original test dataset.
- Report at least RMSE, MAE, and R-squared score.
- Include additional metrics or visualizations if useful, such as predicted vs. actual plots or residual plots.
- Interpret the model performance and identify any limitations.

5. Evidently AI Monitoring Setup (15 Points)

- Set up model monitoring using Evidently AI.
- Create a reference dataset using appropriate baseline data, such as training data or original test data.
- Create monitoring reports for data drift and model performance or regression performance where applicable.
- Generate and save Evidently reports or dashboards, such as HTML files, screenshots, or exported artifacts.
- Explain which monitoring checks were configured and why they are relevant.

6. Modified Test Dataset Creation (10 Points)

- Create a separate modified test dataset for Scenario A.
- Create a separate modified test dataset for Scenario A + B.
- Create a separate modified test dataset for Scenario A + B + C.
- Preserve the original test dataset without overwriting it.
- Validate that medianincome, povertypercent, and avghouseholdsize were changed correctly for each scenario.

7. Scenario-Based Model Scoring and Accuracy Verification (10 Points)

- Run the trained model on the original test dataset and all three modified test datasets.
- Report model accuracy metrics for the original test dataset and each scenario.
- Compare metric changes across Scenario A, Scenario A + B, and Scenario A + B + C.
- Discuss how the controlled input changes affected model predictions and evaluation metrics.

8. Monitoring Analysis for Input and Output Changes (15 Points)

- Run Evidently monitoring for each modified test dataset against the reference dataset.
- Verify whether monitoring detects changes in medianincome, povertypercent, and avghouseholdsize.
- Analyze whether model output or prediction distribution changed across scenarios.
- Include screenshots, HTML reports, or summary tables showing detected drift or monitoring results.
- Explain which changes were detected clearly, which were not detected, and why.

9. Documentation and Professional Practices (5 Points)

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- Provide clear documentation and readable code.
- Organize the repository or notebook so the workflow is easy to reproduce.
- Include comments, markdown explanations, and a summary of results.
- Document operational lessons learned from the monitoring scenarios.

Assignment Requirements

- Use the attached cancer dataset and Data Dictionary.docx.
- Predict target_deathrate using a train/test split methodology.
- Evaluate the baseline model on the original test dataset.
- Use Evidently AI to monitor input data changes and model output changes.
- Create and evaluate three modified test datasets: Scenario A, Scenario A + B, and Scenario A + B + C.
- For each scenario, run the model, report accuracy metrics, and generate monitoring evidence.
- Document preprocessing, assumptions, model choices, monitoring checks, and results.
- Submit code and generated monitoring outputs.

Deliverables

- Jupyter notebook(s) or Python scripts containing the complete workflow
- requirements.txt or environment.yml
- README with setup and execution instructions
- Baseline train/test split and model training code
- Baseline model evaluation metrics and visualizations
- Modified test datasets or code that reproducibly creates them
- Evaluation metrics for original test data and all three modified scenarios
- Evidently AI monitoring reports for original and modified datasets
- Screenshots or exported HTML reports showing input drift and model output changes
- Written discussion of accuracy changes, monitoring findings, and operational implications

Discussion Questions

- How did each controlled input change affect model accuracy?
- Which scenario caused the largest change in prediction behavior?
- Did Evidently detect the intended input changes? Why or why not?
- Did the model output distribution change even when accuracy did not change significantly?
- What monitoring checks would you keep if deploying this model in production?
- What additional safeguards would you add before using this model for real-world decisions?

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Appendix: Data Dictionary

The data was aggregated from a number of sources including the American Community Survey ([census.gov](https://www.census.gov)), clinicaltrials.gov, and cancer.gov. Most of the data preparation process can be viewed [here](#).

TARGET_deathRate: Dependent variable. Mean *per capita* (100,000) cancer mortalities(*a*)

avgAnnCount: Mean number of reported cases of cancer diagnosed annually(*a*)

avgDeathsPerYear: Mean number of reported mortalities due to cancer(*a*)

incidenceRate: Mean *per capita* (100,000) cancer diagnoses(*a*)

medianIncome: Median income per county (*b*)

popEst2015: Population of county (*b*)

povertyPercent: Percent of populace in poverty (*b*)

studyPerCap: *Per capita* number of cancer-related clinical trials per county (*a*)

binnedInc: Median income per capita binned by decile (*b*)

MedianAge: Median age of county residents (*b*)

MedianAgeMale: Median age of male county residents (*b*)

MedianAgeFemale: Median age of female county residents (*b*)

Geography: County name (*b*)

AvgHouseholdSize: Mean household size of county (*b*)

PercentMarried: Percent of county residents who are married (*b*)

PctNoHS18_24: Percent of county residents ages 18-24 highest education attained: less than high school (*b*)

PctHS18_24: Percent of county residents ages 18-24 highest education attained: high school diploma (*b*)

PctSomeCol18_24: Percent of county residents ages 18-24 highest education attained: some college (*b*)

PctBachDeg18_24: Percent of county residents ages 18-24 highest education attained: bachelor's degree (*b*)

PctHS25_Over: Percent of county residents ages 25 and over highest education attained: high school diploma (*b*)

PctBachDeg25_Over: Percent of county residents ages 25 and over highest education attained: bachelor's degree (*b*)

PctEmployed16_Over: Percent of county residents ages 16 and over employed (*b*)

PctUnemployed16_Over: Percent of county residents ages 16 and over unemployed (*b*)

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PctPrivateCoverage: Percent of county residents with private health coverage (*b*)

PctPrivateCoverageAlone: Percent of county residents with private health coverage alone (no public assistance) (*b*)

PctEmpPrivCoverage: Percent of county residents with employee-provided private health coverage (*b*)

PctPublicCoverage: Percent of county residents with government-provided health coverage (*b*)

PctPublicCoverageAlone: Percent of county residents with government-provided health coverage alone (*b*)

PctWhite: Percent of county residents who identify as White (*b*)

PctBlack: Percent of county residents who identify as Black (*b*)

PctAsian: Percent of county residents who identify as Asian (*b*)

PctOtherRace: Percent of county residents who identify in a category which is not White, Black, or Asian (*b*)

PctMarriedHouseholds: Percent of married households (*b*)

BirthRate: Number of live births relative to number of women in county (*b*)

(*a*): years 2010-2016

(*b*): 2013 Census Estimates